

RES.6-012 Introduction to Probability Lectures 11 & 12
Textbook Notes - Section 4.1 & 4.2 04-06-2026

Section 4.1 - Derived Distributions

Architecture

$$\underbrace{Y = g(X)}_{\text{Distribution of an RV}} \xrightarrow{\text{CDF method}} \underbrace{f_Y(y) = f_X(h(y)) \left| \frac{dh}{dy} \right|}_{\text{monotonic formula}} \longrightarrow \underbrace{\text{convolution}}_{Z = X + Y}$$

- ↳ Given the distribution of X , what is the distribution of $Y = g(X)$?
- ↳ CDF method is the systematic tool.

The Two-Step CDF Method

- ↳ Standard procedure for finding the distribution of $Y = g(X)$.
"universal"

Step 1: Find the CDF of Y

$$F_Y(y) = P(g(X) \leq y) = \int_{\{x \mid g(x) \leq y\}} f_X(x) dx$$

Step 2: Differentiate to get PDF

$$f_Y(y) = \frac{dF_Y}{dy}(y).$$

Example: X uniform on $[0, 1]$, $Y = \sqrt{X}$

$$F_Y(y) = P(\sqrt{X} \leq y) = P(X \leq y^2) = y^2$$

$$f_Y(y) = \frac{d}{dy} y^2 = 2y, \quad 0 \leq y \leq 1$$

- ↳ The support maps from $[0, 1]$ to $[0, 1]$

- ↳ Square root preserves the range but reshapes the distribution toward higher values.

• Linear Functions

For $Y = aX + b$ with $a \neq 0$

$$f_Y(y) = \frac{1}{|a|} \cdot f_X\left(\frac{y-b}{a}\right)$$

Intuition $\rightarrow$ stretching the range by $|a|$ requires compressing the density by $1/|a|$ to keep total area = 1

Results $\rightarrow$ Linear function of an Exponential $\rightarrow$ still Exponential, if $(b \neq 0, a > 0)$

$\rightarrow$ Linear function of a Normal $\rightarrow$ Always Normal.

Normal case $\rightarrow Y = aX + b$ gives:

$$f_Y(y) = \frac{1}{\sqrt{2\pi} \cdot |a| \cdot \sigma} \cdot \exp\left\{-\frac{(y - a\mu - b)^2}{2a^2\sigma^2}\right\}$$

$\hookrightarrow$ Normal with mean $a\mu + b$ & variance $a^2\sigma^2$

$\hookrightarrow$ Normality preserved under linear transforms.

• Monotonic Case - The General Formula

$\hookrightarrow$ When g is strictly monotonic over the range of X , the CDF method yields a clean closed-form formula.

Setup: If $y = g(x) \Leftrightarrow x = h(y)$ where h is the inverse of g ,

$$f_Y(y) = f_X(h(y)) \left| \frac{dh}{dy}(y) \right|$$

$\hookrightarrow$ handles increasing / decreasing cases

Example: $Y = \frac{180}{X}$, X uniform $[30, 60]$

$$\therefore h(y) = \frac{180}{y} \quad \frac{dh}{dy} = -\frac{180}{y^2}$$

$$\therefore f_Y(y) = \frac{1}{30} \cdot \frac{180}{y^2} = \frac{6}{y^2}, \quad y \in [3, 6]$$

- The non-monotonic case $\rightarrow$ When g is not monotonic
 e.g. $Y = X^2$ where X takes positive & negative values.
 $\hookrightarrow$ Cannot apply General Formula
 $\hookrightarrow$ Return to Standard CDF method & split the problem at the non-monotonic points.

• Functions of Two Random Variables:

CDF method extends to $Z = g(X, Y)$.

$$F_Z(z) = P(g(X, Y) \leq z) = \iint_{\{(x, y) | g(x, y) \leq z\}} f_{X, Y}(x, y) dx dy$$

$\hookrightarrow$ May be difficult to identify correct integration region for each value of z

Maximum Example: $Z = \max(X, Y)$ where X, Y independent, uniform on $[0, 1]$.

$$F_Z(z) = P(X \leq z, Y \leq z) = P(X \leq z) \cdot P(Y \leq z) = z^2$$

$$f_Z(z) = 2z, \quad z \in [0, 1]$$

$\hookrightarrow$ Why it works: maximum exceeds z only if both RVs exceed z .
 Independence factorises the joint probability.

• Convolution - Sums of Independent RVs

For $Z = X + Y$ with X, Y independent:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

Key step in derivation: $P(Z \leq x | X = x) = P(Y \leq z-x)$

{ Differentiating both sides ~~with~~ over $z \rightarrow f_{Z|X}(z|x) = f_Y(z-x)$
 Integrate over $x \rightarrow$ yields convolution }

→ Convolution Results:

• Uniform + Uniform: → X, Y independent uniform on $[0, 1]$, $Z = X + Y$
↳ has a triangular PDF on $[0, 2]$
↳ cannot be guessed from individual distr. alone

• Normal + Normal: → $X \sim N(\mu_x, \sigma_x^2)$, $Y \sim N(\mu_y, \sigma_y^2)$ independent
↳ $Z = X + Y \sim N(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2)$
↳ Sum of independent Normals is Normal.

• Graphical convolution: → For complex limits, visualise by fixing z , flipping f_x to $f_x(z-x)$, sliding it along, and integrating the overlap at each z .
(textbook page 216 - Figure 4.10).

• Section 4.2 - Covariance & Correlation

• Architecture

$$\text{cov}(X, Y) = E[XY] - E[X]E[Y] \xrightarrow{\text{normalise}} \rho(X, Y) \in [-1, 1] \xrightarrow{\text{application}} \text{var}(X+Y) = \text{var}(X) + \text{var}(Y) + \text{covar}(XY)$$

↳ Quantifies the dependence between two RVs → essential for computing variance of sums when independence doesn't hold.

• Covariance

$$\left\{ \text{cov}(X, Y) = E \left[\underbrace{(X - E[X])(Y - E[Y])}_{\text{"the product"}} \right] \right\}$$

↳ The Product is positive when both RVs deviate in the same direction, negative when they deviate oppositely.

↳ Expectation of this product captures the average tendency for joint deviation.

Computational Form :

$$\text{cov}(X, Y) = E[XY] - E[X]E[Y]$$

↳ Special case $\Rightarrow$ ~~cov~~ $\text{cov}(X, X) = \text{var}(X)$.

→ Key Properties

→ Symmetry

$$\text{cov}(X, Y) = \text{cov}(Y, X)$$

→ Scaling

$$\text{cov}(aX, bY) = ab \cdot \text{cov}(X, Y)$$

→ Bilinearity (distributes over sums)

$$\text{cov}(X+Y, Z) = \text{cov}(X, Z) + \text{cov}(Y, Z)$$

→ Independence implies zero covariance

$$X, Y \text{ independent} \Rightarrow \text{cov}(X, Y) = 0$$

↳ NB!!! $\Rightarrow$ converse fails

$$\text{cov}(X, Y) = 0 \neq X, Y \text{ independent}$$

- Zero Covariance only rules out linear dependence.
 ↳ Non-linear dependence can coexist with zero covariance.

• Variance of a Sum - Central Application

$$\text{var}(X+Y) = \text{var}(X) + \text{var}(Y) + 2\text{cov}(X, Y)$$

For n RVs:

$$\text{var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{var}(X_i) + 2 \sum_{\{i,j\} | i < j} \text{cov}(X_i, X_j)$$

↳ When all pairs are independent → cov() term vanishes → retain simple Σ .

↳ When dependent, covariance term either inflate ($\text{cov} > 0$) or deflate ($\text{cov} < 0$) the total variance.

Example: Hat Problem revisited

$X = X_1 + \dots + X_n$, each X_i is Bernoulli ($\frac{1}{n}$)

↳ X_i are not independent, so variance formula fails.

$$\begin{aligned} \text{Covariance: } \text{cov}(X_i, X_j) &= E[X_i X_j] - E[X_i]E[X_j] \\ &= \frac{1}{n(n-1)} - \frac{1}{n^2} = \frac{1}{n^2(n-1)} \end{aligned}$$

$$\text{var}(X) = n \cdot \frac{1}{n} \left(1 - \frac{1}{n}\right) + n(n-1) \cdot \frac{1}{n^2(n-1)} = 1$$

↳ Variance equals 1 for all n .

↳ Linearity of expectation alone could not determine.

• Correlation Coefficient

Covariance has a scaling problem $\rightarrow$ $\text{cov}()$ magnitude depends on the units of X and Y , making comparison across problems meaningless.

Fix:

$$\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X) \text{var}(Y)}}$$

Properties

- Always $-1 \leq \rho \leq 1$
- $\rho = 1$ $\rightarrow$ perfect positive linear relationship:
 $Y - E[Y] = c(X - E[X])$ for some $c > 0$.
- $\rho = -1$ $\rightarrow$ perfect negative linear relationship.
- $\rho = 0$ $\rightarrow$ no linear relationship (possibly nonlinear dependence)
- Dimensionless $\rightarrow$ comparable across problems / no unit

Example: Heads & Tails

X = number of heads ; Y = number of tails in n coin tosses

$$X + Y = n \quad \text{so} \quad X - E[X] = -(Y - E[Y])$$

$$\hookrightarrow \text{gives } \text{cov}(X, Y) = -\text{var}(X)$$

$$\hookrightarrow \rho(X, Y) = \frac{-\text{var}(X)}{\sqrt{\text{var}(X) \cdot \text{var}(X)}} = -1$$

$\hookrightarrow$ Knowing number of heads perfectly determines number of tails.

$\hookrightarrow$ Perfect negative (-1) correlation.

• Finance Connection

Portfolio variance for weights w_1, w_2 :

$$\text{var}(w_1 X + w_2 Y) = w_1^2 \text{var}(X) + w_2^2 \text{var}(Y) + 2w_1 w_2 \text{cov}(X, Y)$$

When $\text{cov}(X, Y) < 0 \rightarrow$ assets move in opposite directions.

$\rightarrow$ Diversification $\rightarrow$ cross term reduces portfolio variance below weighted sum of variances
 $\rightarrow$ mix negatively correlated assets to reduce risk.

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Textbook Notes - Sections 4.3 to 4.6 (exclude 4.4)

Architecture

$$\underbrace{E[X|Y]}_{\substack{\text{4.3: conditional expectation} \\ \text{as a RV}}} \xrightarrow{\text{iterated expansions}} \underbrace{E[E[X|Y]] = E[X]}_{\text{law of iterated expectations}} \xrightarrow{\text{variance version}} \underbrace{\text{var}(X) = E[\text{var}(X|Y)] + \text{var}(E[X|Y])}_{\text{law of total variance}}$$

$$\underbrace{M_X(s) = E[e^{sX}]}_{\text{4.4: moment generating function}} \xrightarrow{\text{independence}} \underbrace{M_{X+Y}(s) = M_X(s) \cdot M_Y(s)}_{\text{addition} \leftrightarrow \text{multiplication}} \xrightarrow{\text{random sums}} \underbrace{M_Y(s) = M_N(\log M_X(s))}_{\text{4.5: } Y = X_1 + \dots + X_N}$$

Section 4.3 Conditional Expectation & Variance Revisited.

$E[X|Y]$ as a Random Variable

- ↳ The conditional expectation $E[X|Y=y]$ is a number that depends on y .
- ↳ Key conceptual shift: define the random variable $E[X|Y]$ as the function of Y whose value is $E[X|Y=y]$ whenever Y takes the value y .

→ $E[X|Y]$ a function of Y → distribution is inherited from the distribution of Y → has ~~an~~^{own} expectation & variance.

Example 8

- A biased coin has probability of heads Y , where Y itself is random with a known distribution over $[0, 1]$.
- Coin is tossed n times and X counts the number of heads.
- For any $y \in [0, 1]$, $E[X|Y=y] = ny$
- Therefore $E[X|Y] = nY$ (random variable determined entirely by Y).

• Law of Iterated Expectations

Since $E[X|Y]$ is a RV, its expectation ~~and~~ is computable:

$$\underbrace{E[E[X|Y]]} = \begin{cases} \sum_y E[X|Y=y] \cdot p_Y(y) & , \quad Y \text{ discrete} \\ \int_{-\infty}^{\infty} E[X|Y=y] \cdot f_Y(y) dy & , \quad Y \text{ continuous} \end{cases}$$

↳ From total expectation theorem

Results: $E[E[X|Y]] = E[X]$

Interpretation: • First compute $E[X|Y=y]$ for each y .
• Next average over Y .

↳ Law of iterated expectations (tower property)

Example:

- A stick of length ℓ is broken at a uniformly chosen point.
- Left piece is kept.
- Process repeats on the remaining piece.

→ Let Y = length after first break, X = length after second break
→ Then $E[X|Y] = Y/2$ (uniform breakpoint on a piece of length Y)
and $E[Y] = \ell/2$
→ By iterated expectations:
$$E[X] = E[E[X|Y]] = E\left[\frac{Y}{2}\right] = \frac{1}{2}E[Y] = \frac{\ell}{4}$$

• Forecast Revision Property:

↳ If $E[X]$ is an initial forecast and $E[X|Y]$ is a revised forecast after observing Y , then $E[E[X|Y] - E[X]] = 0$

↳ The expected revision is zero, if the expected revision were nonzero, the original forecast should have been adjusted.

Property: $E[X g(Y) | Y] = g(Y) \cdot E[X | Y]$ } → Given value of Y , $g(Y)$ is a constant, and can be pulled outside

• Conditional Expectation as an Estimator:

Define $\hat{X} = E[X|Y]$ as an estimator of X given the observation Y .

Estimation error is $\tilde{X} = X - \hat{X}$

Properties of Estimation Error:

→ Zero conditional mean

$$E[\tilde{X}|Y] = E[X|Y] - E[\hat{X}|Y] = E[X|Y] - E[X|Y] = 0$$

↳ Second equality holds because $\hat{X}$ is a function of Y , so

$$E[\hat{X}|Y] = \hat{X} = E[X|Y]$$

↳ By iterated expectations, unconditional mean $E[\tilde{X}] = 0$.

→ Uncorrelated with the Estimate

$$E[\hat{X}\tilde{X}] = E[E[\hat{X}\tilde{X}|Y]] = E[\hat{X} \cdot E[\tilde{X}|Y]] = E[\hat{X} \cdot 0] = 0$$

$$\therefore \text{cov}(\hat{X}, \tilde{X}) = E[\hat{X}\tilde{X}] - E[\hat{X}]E[\tilde{X}] = 0.$$

Since $X = \hat{X} + \tilde{X}$ and the two terms are uncorrelated:

$$\text{var}(X) = \text{var}(\hat{X}) + \text{var}(\tilde{X})$$

The total variability in X → 1. explained by Y (through $\hat{X}$)
→ 2. residual variability (through $\tilde{X}$)

• Law of Total Variance

Define the conditional variance as the random variable $\text{var}(X|Y)$ whose value is $\text{var}(X|Y=y)$ when $Y=y$.

→ Using $E[\tilde{X}] = 0$ & Law of Iterated Expectations:

$$\text{var}(\tilde{X}) = E[\tilde{X}^2] = E[E[\tilde{X}^2|Y]] = E[\text{var}(X|Y)]$$

→ Substituting into: $\text{var}(X) = \text{var}(\hat{X}) + \text{var}(\tilde{X})$:

$$\boxed{\text{var}(X) = E[\text{var}(X|Y)] + \text{var}(E[X|Y])}$$

① → average variability with each conditioned sub-problem.

② → variability between the conditional means.

②

Example:

With $Y \sim \text{Uniform}[0, 1]$ as the probability of heads and
 $X = \text{number of heads in } n \text{ tosses:}$

$$E[X|Y] = nY \quad \& \quad \text{var}(X|Y) = nY(1-Y).$$

Then:

$$\begin{aligned} E[\text{var}(X|Y)] &= E[nY(1-Y)] = n(E[Y] - E[Y^2]) \\ &= n\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{n}{6} \end{aligned}$$

$$\text{var}(E[X|Y]) = \text{var}(nY) = n^2 \cdot \frac{1}{12} = \frac{n^2}{12}$$

$$\text{var}(X) = \frac{n}{6} + \frac{n^2}{12}$$

~~Section 4.4 Transforms~~

& Sect 10.

• Section 4.5 Sum of a Random Number of Independent RVs.

• Setup

Consider the sum $Y = X_1 + X_2 + \dots + X_N$ where:

- $N \rightarrow$ RV taking nonnegative integer value (if $N=0$, then $Y=0$)
- $X_1, X_2, \dots \rightarrow$ identically distributed with common mean $E[X]$ & $\text{var}(X)$
- $N, X_1, X_2, \dots$ are all mutually independent.

(Strategy) $\rightarrow$ condition on $\{N=n\}$ to reduce to familiar fixed sum case.
 $\rightarrow$ use laws of iterated expectations &
 $\hookrightarrow$ total variance to remove condition.

• Mean of a Random Sum

For a fixed n : $E[Y | N=n] = E[X_1 + \dots + X_n] = nE[X]$

$\hookrightarrow$ Since it holds for every n , the RV $E[Y | N] = N \cdot E[X]$

$\hookrightarrow$ By law of iterated expectations:

$$E[Y] = E[N] \cdot E[X]$$

• Variance of a Random Sum

For a fixed n : $\text{var}(Y | N=n) = n \cdot \text{var}(X)$ (indep of X_i).

$\hookrightarrow$ So: $\text{var}(Y | N) = N \cdot \text{var}(X)$

$\hookrightarrow$ By law of total variance:

$$\text{var}(Y) = E[\text{var}(Y | N)] + \text{var}(E[Y | N]) = E[N] \cdot \text{var}(X) + (E[X])^2 \text{var}(N)$$

$$\text{var}(Y) = E[N] \cdot \text{var}(X) + (E[X])^2 \text{var}(N)$$

① $\rightarrow$ uncertainty from variability within each X_i , scaled by expected # items.

② $\rightarrow$ uncertainty from not knowing how many terms appear, scaled by squared mean contribution of each term.

• Transform of a Random Sum

Conditioned on $N=n$, Y is sum of n independent RVs:

$$E[e^{sY} | N=n] = (M_X(s))^n$$

Using iterated expectations

$$M_Y(s) = E[(M_X(s))^N] = \sum_{n=0}^{\infty} (M_X(s))^n \cdot p_N(n)$$

...

$$M_Y(s) = M_N(\log M_X(s))$$

• Worked Examples

Gas Station p. 242

Three gas stations, each open with prob $\frac{1}{2}$ independently.

Gas available at each is Uniform $(0, 1000)$.

Number open is $N \sim \text{Binomial}(3, \frac{1}{2})$ with transform $M_N(s) = \left(\frac{1+e^s}{2}\right)^3$

The gas at each station has $M_X(s) = \frac{e^{1000s}-1}{1000s}$.

Total gas available has transform:

$$M_Y(s) = \left(\frac{1}{2} \left(1 + \frac{e^{1000s}-1}{1000s}\right)\right)^3$$

Geometric number of Exponentials p. 243

Jane visits bookstores (geometrically distributed number N with parameter p) and spends exponentially distributed time (parameter λ) at each.

Total time $Y = X_1 + \dots + X_N$ has:

$$E[Y] = \frac{1}{p} \cdot \frac{1}{\lambda} = \frac{1}{p\lambda}$$

$$\text{var}(Y) = \frac{1}{p\lambda^2} + \frac{1}{\lambda^2} \cdot \frac{1-p}{p^2} = \frac{1}{\lambda^2 p^2}$$

For the transform: $M_N(s) = \frac{pe^s}{1-(1-p)e^s}$, $M_X(s) = \frac{\lambda}{\lambda-s}$

↳ Replacing e^s with $M_X(s)$:

$$M_Y(s) = \frac{p\lambda/(\lambda-s)}{1-(1-p)\lambda/(\lambda-s)} = \frac{p\lambda}{p\lambda-s}$$

↳ Transform of Exponential with parameter $p\lambda$, so $Y \sim \text{Exponential}(p\lambda)$

Result: sum of a fixed num of independent exponentials is Erlang not Exponential
 sum of a geometric num of exponentials is Exponential
 ↳ randomness of N "restores" memoryless property